

DRAFT ESSAY · NOT SUBMITTED · NOT REFEREED

Every process function lifts to a unitary process

A full note on the two failures, the bookkeeping, and the cancellation.

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What is new, in ordinary language.

Three statements are proved. They are not the same statement.

First, the open map. Add an extra input that nobody chose, and keep a copy of what was sent. The map that hides the process under the extra input and stores the output in the copy can be run forwards and backwards, for every process, even the ones that do not finish uniquely. That is a bijection. It does not feed the output back in as the next input.

Second, the closed map, and only for processes that finish every choice once. If two possible outputs differ, some participant wrote different things and received the same reply. At that desk the two contributions cancel, whichever desk it is. So the closed map is unitary. This is the new step. The 2019 paper had the open map and left this open.

Third, the closed map is not unitary for every process. The process that just copies its output to its input has 2^n consistent readings where a good map would have one. On three people that is 8. So “closed” and “open” are different claims, and the first is false without the unique-finish rule.

Checked in Lean 4.34.1, with no unfinished steps: the witness at any party, both directions of the open map, and the cancellation for integer matrices. Not proved: a spacetime for the three-person example. A metric is a way of realizing the rule, not part of the rule.

Abstract

A time-travel story can fail in two simple ways. The traveller tries to prevent the trip, and no version of the story fits. Or the story fits in more than one way, and nothing picks which one. Philosophers have usually answered by saying that the bad stories do not happen, or that “can” means something different once the whole past is in view [1], [2]. This essay uses a sharper rule. A process is allowed only when every choice the people in it might make finishes the story exactly once. No finish is the grandfather case. Two finishes are the note that writes itself.

There is a reversible way to write any such process down. Add a source, an extra input nobody chose, and a sink, a copy of what was sent. The map that hides the process under the source and stores the output in the sink can be run forwards and backwards, for every process, even the inconsistent ones [3]. That is the open case. It does not yet feed the output back in as the next input. Doing that is the closed case, and it is not well behaved for every process. The process that just copies its output to its input has eight consistent readings on three people, where a good map would have one.

What the unique-finish rule adds is the cancellation. If two possible outputs differ, some participant wrote different things and received the same reply. At that desk the two contributions cancel. So the closed map is reversible precisely when every choice already finished once. A 2026 note gives two sufficient conditions for a related claim; neither is used here [4]. The essay does not build a spacetime for the example. It does not need one: the rule is a condition on messages, and a metric would be a way of realizing the rule, not part of the rule. It does not say whether the traveller was free.

1. How the argument developed

The literature did not begin with a matrix. It began with a word, then with a demand about interventions, then with a differential equation, then with three quantum filters, and only then with a map that finishes every local choice once. The proof in this essay is easier to follow if that order is kept. The grandfather story and the note in the coat are the two failures. Lewis and Novikov are earlier answers to those failures [1], [2]. Deutsch's fixed point and post-selection are later answers of the same shape [15], [19]. A process function is a different shape. It refuses the map, rather than the history.

The bookkeeping step came next, in 2019, and it does not need the unique-finish rule [3]. A source and a sink turn any function into a bijection. Closing the output onto the input is a further step, and that step fails for the copy rule. The cancellation, which is the new step, uses the unique-finish rule to kill the extra reading. The 2026 note of Daher Ahmed and Kunjwal sits beside this argument, not under it [4]. It gives two sufficient conditions for a unitary to be a process-function unitary. This essay does not use those conditions, and it does not claim they are necessary.

2. Two ways a loop can fail

The cheapest objection to a cause that comes after its effect is a rule about the word. A cause, it is said, is what comes before, so an effect that precedes its cause is a contradiction in terms. Dummett's

1954 paper to the Aristotelian Society, and the 1964 essay “Bringing about the Past,” treat that as a stipulation about English, not a discovery about the world [5], [6]. One can reserve the word cause for the earlier relatum and still ask whether a later event is necessary for an earlier one. Faye’s Stanford entry still takes the temporal order to be contingent, and still treats the consistency puzzles as unresolved [7].

The objection that survives the lexical move is Black’s, in *Analysis* in 1956 [8]. Suppose an earlier event E is supposed to be the effect of a later event C . An agent who observes E can, it seems, intervene so that C does not occur. If the intervention succeeds, C was not necessary for E . If it cannot succeed, the agent lacks an ability that the local facts appear to leave open, and that lack wants an explanation that is not the fiat that the past is fixed. Horwich later pressed the same point as a constraint on any theory that lets later events bear on earlier ones: the theory must say what happens to the intervention, not merely announce that paradoxical interventions do not arise [9].

Lewis’s 1976 paper is the standard reply, and it is a reply about modality, not about dynamics [1]. One can kill one’s grandfather, relative to the local facts and the laws in a small region. One cannot, relative to the whole past, because the past already contains the failure. The two cans are not in conflict once their contexts are distinguished. That dissolves a verbal puzzle. It does not say which global histories the laws admit, and it does not say why the local ability fails to compose with the rest of the world. A reader who wants a criterion, rather than a reminder that modal contexts shift, is entitled to remain unsatisfied. The slogan that only consistent loops occur has the same shape. Until consistency is a property of the admitted maps, the slogan is a filter applied after the solutions have been written down.

THE CLERK AND THE CLOSED WINDOW

The telegraph office closed at six. Mara knew this because the telegram that had sent her there said so, and because she had watched the clerk lock the grille at 17:58 on the afternoon she was trying to prevent. The message named a train, a platform, and a man who would not be on it if the message were never sent. She had the coins. She had the form. What she did not have was a window that would take the form.

She tried the obvious repairs. She arrived at five, and the clerk was already locking up, citing a wire from the morning that the office would close early. She arrived at four, and the wire was from the day before. She stood in the doorway and did nothing, and a boy she did not recognise handed the clerk a slip in her handwriting, which she had not written, and which the clerk filed under the heading she had come to erase. None of these is a story about fate. Each is a story about a function with no fixed point. The intervention — send, do not send, send a different text — asks the office to

return an output that the office, given that intervention, does not return. Lewis would say she can, locally, and cannot, globally [1]. The clerk says the window is closed. The interesting question is which maps from her action to the clerk's action are such that every action she might take completes exactly once. The office in the story is not such a map. That is why the story cannot be finished.

THE NOTE IN THE COAT

The second story is the one people tell when they want the loop to be charming. A woman finds a note in the pocket of a coat she does not remember buying. The note is in her hand, and it tells her which coat to buy. She buys the coat. She copies the note, because the original has faded, and she leaves the copy in the pocket for the woman who will find it, who is herself. Two details are usually omitted. She might have written a different sentence. And the faded note and the copy do not say quite the same thing, yet both, if taken as the instruction, lead her to a coat that contains a note.

Call the sentence the output and the coat the input. The charming story is a map with two fixed points: the faded wording, and the copy. A third wording, in which she writes nothing, has none. The bootstrap is not a paradox of information appearing from nowhere, though it is often sold as one. It is a failure of uniqueness. The grandfather story was a failure of existence. A process function refuses both, and for the same reason. If two global strings are each fixed by one intervention, that intervention is not an admissible local choice for that map. The note may still be written. It may not be written in two ways at once by a map we are calling a process.

3. What the spacetime theories did with the same two failures

General relativity makes the question a problem in differential equations. Gödel's 1949 solution of Einstein's equation has closed timelike curves through every event [10]. The solution is a curiosity of global structure, not a laboratory, but it removes the comfort that the field equation itself forbids loops. Morris, Thorne and Yurtsever, in 1988, showed something sharper for people who want a machine rather than a cosmology [11]. A traversable wormhole, held open by stress-energy that violates the averaged null energy condition, can be converted into a time machine by relative motion of the mouths. The conversion is a construction in classical general relativity plus an exotic-matter assumption. It is not a theorem that such matter exists. It is a theorem that if it does, the Cauchy problem changes character.

Friedman, Morris, Novikov, Echeverria, Klinkhammer, Thorne and Yurtsever stated the changed problem in 1990 [2]. For a spacetime that develops closed timelike curves, the initial-value problem on a partial Cauchy surface is not the ordinary one. Data that would be admissible in a globally

hyperbolic spacetime can fail to extend, or can extend in more than one way, once the curve lets a system interact with its own past. Their proposal, Novikov's self-consistency principle, is that only globally self-consistent solutions occur. Inconsistent data are assigned measure zero. Nothing reaches in to stop the traveller. The contradictory solutions are simply not among the solutions.

The billiard makes the underdetermination explicit. Echeverria, Klinkhammer and Thorne, in 1991, integrated a hard-sphere ball in a wormhole spacetime that takes the ball backward in time [12]. Some initial trajectories have one self-consistent continuation. Some have two. Some appear to have none. The multiplicity is a property of the classical initial-value problem, not an artefact of quantum indeterminacy. Carlini, Frolov, Mensky, Novikov and Soleng later derived a version of self-consistency from stationarity of the action, and found the same ill-posedness: the stationary points do not pick a unique extension [13]. The fifth author of that paper is H. H. Soleng, not Solodukhin. The journal is the *International Journal of Modern Physics D* 4, 557–580 (1995), with an erratum in volume 5, page 99 (1996). Earman's 1995 survey is the philosophical summary that still holds [14]. A consistency condition imposed on the solution set after the dynamics have been solved is not itself a dynamics. It is a selection rule. Selection rules can be true. They are not explanations of why the discarded solutions were not solutions of the equations.

That is the classical shape of the gap. A logic of loops has to block grandfather cases and bootstrap cases without collapsing into the stipulation that only consistent loops occur. The stipulation is what Novikov's principle is. The billiard shows why a stipulation about measure does not finish the job: among the consistent solutions, the classical theory often does not choose.

4. Three quantum replies

Deutsch's 1991 model is the first quantum reply that became standard [15]. A system travelling on the closed curve is described by a density operator fixed by the partial trace of its interaction with the chronology-respecting systems. In finite dimension a fixed point exists, so no particular action is forbidden. The grandfather paradox becomes a mixed state. The agent who sets out to kill an earlier self is correlated with a branch in which the attempt fails, and the reduced state on the curve is consistent. The cost is nonlinearity of the evolution on the chronology-respecting systems. The model can clone unknown states. Aaronson and Watrous showed in 2009 that the computational power is PSPACE: classical and quantum computing with these curves are equivalent, and both are as strong as polynomial space [16]. Bennett, Leung, Smith and Smolin argued the same year that the power is an artefact [17]. The fixed-point condition is imposed on a distribution that an ordinary preparation, tensored with a maximally mixed state on the curve, would not produce. Tolkdorf and Verch showed that the Deutsch condition can be met to arbitrary precision by a quantum field

theory on a globally hyperbolic spacetime, with no closed curve at all [18]. A Deutsch fixed point is therefore not diagnostic of a loop. It is a nonlinear closure condition that can be mimicked.

Lloyd, Maccone and collaborators replace the fixed point by post-selection [19]. In the post-selected model, inconsistent branches are the ones that fail the post-selection. The grandfather paradox is resolved by the experiment not happening. Process functions sit inside this class in a precise sense: the event list of a consistent classical process appears as the record conditional on the post-selection succeeding. That is a simulation. It is not an account of why the inconsistent branch was never a physical option. Post-selection is a filter on samples, of the same shape as a measure-zero axiom. It tells you which runs to keep. It does not tell you which maps were admissible before the runs were generated.

The third reply is the process matrix of Oreshkov, Costa and Brukner, and the process function that sits underneath it [20]. A process matrix is a channel from the outputs of several parties to their inputs, with no assumption that the parties are ordered in time. A process function is the deterministic skeleton: a map w from output strings to input strings such that every choice of local deterministic interventions has exactly one fixed point. The parties may be in a definite causal order, or not. The definition does not say. What it says is that whatever local choice each party makes, the loop completes, and completes once. Grandfather interventions are excluded because they have no completion. Bootstrap interventions are excluded because they have more than one. The exclusion is a property of the map, not a filter applied to a list of histories after the map has been evaluated.

5. The rule, written as an equation

Let each party have a finite alphabet. The case that matters here is a bit, with addition written as XOR. A local intervention at party k is a function f_k from that party's input bit to its output bit. A global intervention is a tuple $f = (f_k)$. A process function is a map

$$w: \{0, 1\}^n \rightarrow \{0, 1\}^n$$

such that for every f there is exactly one string o with

$$o_k = f_k(w_k(o)) \quad \text{for all } k. \tag{1}$$

The string o is the output. The string $w(o)$ is the input the process returns. The equation says that each party's intervention, applied to the input it received, reproduces the output it sent. Unique solvability is the whole definition. There is no further clause about probability, and no clause about a metric.

Two maps fail the definition for opposite reasons, and both failures will matter. The identity loop $w(o) = o$ is fixed by every intervention that is the identity, and also by nothing, or by two strings, according to the choice of f . Concretely, the local function that sends 0 to 1 and 1 to 0 has no fixed point on the identity loop, and the identity intervention fixes every string. The swap on two bits, $w(o_0, o_1) = (o_1, o_0)$, is fixed by negation on each side at both $(0, 1)$ and $(1, 0)$. A process function cannot have that.

The Lugano function is the example that keeps the definition from being vacuous. On three bits it may be written so that each party's input is a function of the other parties' outputs, and every local choice completes once. It is not a causal order. No party is in the past of the others in a way that would let one simulate the map by a circuit without a loop. Araújo, Feix, Navascués and Brukner, in the 2017 purification postulate, treat a map of this kind [21]. Vanrietvelde, Kristjánsson and Barrett gave the routed-circuit framework in which such a map can be drawn [22]. Nick Ormrod is not an author of that paper, and the paper does not reconstruct Lugano. Those are existence proofs and a drawing language for one function, or for a class of circuits. The question left open in 2019 was whether every process function has a reversible extension that quantizes.

6. The bookkeeping step, which works for every rule

Baumeler, Costa, Ralph, Wolf and Zych, in 2019, proved that every process function extends to a bijection by the addition of a source and a sink [3]. The construction is short, and it does not use the unique-finish rule. Add a source string e and keep the output string o as a second factor. Define

$$U(o, e) = (w(o) \oplus e, o) \quad (2)$$

and

$$U^{-1}(i, s) = (s, w(s) \oplus i). \quad (3)$$

This is a permutation of the computational basis for any function w , process function or not. On bits the translation $e \mapsto a \oplus e$ is an involution,

$$a \oplus (a \oplus b) = b, \quad (4)$$

so the inverse does not need a separate subtraction.

Left inverse. Start from (o, e) . The image is $(w(o) \oplus e, o)$. Apply U^{-1} . The first component is the second component of the image, which is o . The second component is $w(o) \oplus (w(o) \oplus e) = e$. So $U^{-1} \circ U$ is the identity. Right inverse. Start from (i, s) . The image is $(s, w(s) \oplus i)$. Apply U . The second component is s . The first component is $w(s) \oplus (w(s) \oplus i) = i$. So $U \circ U^{-1}$ is the identity. Both compositions are the identity for every w . Fixing the source at 0 recovers w , and that recovery is

not uniqueness of the fixed point. The identity loop and the swap have this bijection and are not process functions.

The same two lines are the proof for an arbitrary finite number of parties. XOR cancels coordinatewise, and the sink is copied through unchanged. This is the open case. Nothing has been fed back. Each party may then apply a unitary on its input bit plus an ancilla prepared in $|0\rangle$. Columns of a unitary indexed by ancilla input 0 are a subset of the columns, so they are orthonormal. An isometry after a unitary is an isometry. That sentence is the open instrument step. It is not the closed contraction.

The 2019 paper showed the extension and said the quantum step was open. It was unclear whether every classical deterministic process function lifts to a valid unitary quantum process. They checked one finite-dimensional case. Daher Ahmed and Kunjwal, in a note of 30 September 2026, give two sufficient conditions, mutual exclusivity of control conditions and unambiguity, for a unitary to be a process-function unitary [4]. Neither sufficient condition is necessary for the cancellation below. Lugano sits outside that subclass and was already known to work by another construction. The argument below does not use those conditions. The dilation is the source-and-sink permutation, and the process-function clause is the cancellation.

7. Closing the loop is a different step

Closing the output onto the input is the further demand. Against local identity tables the closed amplitude is

$$J_{s,e} = \prod_k \mathbf{1}[s_k = w_k(s) \oplus e_k]. \quad (5)$$

For the copy rule $w(s) = s$ this equals 1 for every output string when the source is 0, and equals 0 otherwise. The diagonal entry is the number of output strings,

$$(J^\dagger J)_{0,0} = 2^n. \quad (6)$$

On three people that is 8, not 1. The same count is 2 on one person. For the two-bit swap it is 2 at sources 00 and 11. So $J^\dagger J = I$ fails for maps that are not process functions. A witness can still kill an off-diagonal of a process function. Universality cannot. The column-slice inheritance of the open step does not cancel this. The open shuffle has not fed the output back. Closing it is a different map.

This is the control the proof needs. A check that only expands $MM^\dagger$ for a process function can look more general than it is. The copy rule satisfies the bookkeeping identity and fails the unique-finish rule, and its gram is not the identity. The open case and the closed case had been written as one sentence. They are not one sentence.

8. The witness, which is the unique-finish rule rearranged

The combinatorial fact that makes the contraction work is the definition, rearranged.

Lemma. Let w be a process function. If $s \neq s'$, then there is a party k with
$$s_k \neq s'_k \text{ and } w_k(s) = w_k(s'). \quad \text{\label{eq:witness}}$$

Proof. Suppose not. Then, whenever $w_k(s) = w_k(s')$, already $s_k = s'_k$. Define a local function by sending $w_k(s)$ to s_k , sending $w_k(s')$ to s'_k when that input differs, and sending everything else to 0.

The two clauses agree wherever the inputs agree, by the supposition, so the choice is a function.

Both s and s' are fixed points of that choice. A process function cannot have two. Hence some party is a witness of the displayed kind.

The witness need not be the first party. On the Lugano function the witness of an unequal pair sits at index 0, 1, or 2. A cancellation proved only for the head party does not cover the function that motivated the question. The swap is the witness in the other direction: (0, 1) and (1, 0) agree in w at both positions and disagree as strings, so the local negation fixes both. The identity loop fails for the same reason. Both are already known not to be process functions. The lemma says why, in a form the contraction can use.

The note in the coat, told the other way, is this lemma. Suppose the faded note and the copy differ in the first word and agree in the second. Then there is a party — the first word — at which the two strings disagree as outputs and agree as inputs. That party is the witness. The contraction will put the inner product of two distinct rows at exactly that position, and the product will vanish. The story does not prove this. It is only a reason to keep the witness in view when the indices start.

9. The contraction

U is a permutation unitary for any w . Validity uses the lemma. Let each party carry a local matrix V_k , with rows indexed by the sink bit and columns by the input bit seen at that party. The source-to-sink matrix element is the contraction of U against these matrices. Because U has a single 1 in each column, the sum over the intermediate output and the intermediate input collapses. The surviving term has output s and input $w(s) \oplus e$, and the amplitude is a product because each V_k acts on its own factor:

$$M_{s,e} = \prod_k \langle s_k | V_k | w_k(s) \oplus e_k \rangle. \quad (7)$$

No process-function hypothesis has been used. The same formula holds for the identity loop and for the swap. They will fail later, at the off-diagonal, which is where they should fail.

The gram entry is the sum over the source,

$$(MM^\dagger)_{s,s'} = \sum_e M_{s,e} \overline{M_{s',e}}. \quad (8)$$

The bar is complex conjugation. In the integer file the entries are integers and the bar is the identity. An involution on the second factor is not moved by the reindexing that follows.

10. The sum of a product

The first algebraic step does not know what w is. It is the statement that a sum of a product over a product group is the product of the sums. Write $A_k: \{0, 1\} \rightarrow R$ for the local factor at party k , with R a commutative ring, and write $\sum_{e \in \{0,1\}^n} \prod_k A_k(e_k)$

11. The change of variables

For each party, set $i_k = w_k(s) \oplus e_k$. Translation by a fixed bit is a bijection of $\{0, 1\}$, and on bits it is an involution, by (3) in the open-case section, the same XOR identity. The two-point sum is therefore invariant:

$$\sum_{e \in \{0,1\}} g(a \oplus e) = \sum_{i \in \{0,1\}} g(i). \quad (9)$$

The case $a = 0$ is the identity. The case $a = 1$ swaps the two summands, and addition commutes. Substituting into the factored gram puts the shift

$$\delta_k = w_k(s') \oplus w_k(s) \quad (10)$$

on the second factor only. Conjugation, if it is present, sits inside that factor and is not reindexed. The identification is

$$(MM^\dagger)_{s,s'} = \prod_k \sum_{i \in \{0,1\}} \langle s_k | V_k | i \rangle \overline{\langle s'_k | V_k | i \oplus \delta_k \rangle}. \quad (11)$$

It holds for every finite party list, every map w , and every family of integer matrices. The process-function clause has still not been used. That is intentional. The identification is kinematics. Unitarity is the clause.

12. Unitarity, with the witness anywhere

Assume now that each V_k has orthonormal rows:

$$\sum_i \langle r|V_k|i\rangle \overline{\langle r'|V_k|i\rangle} = \delta_{r,r'}. \quad (12)$$

A zero shift then makes the k -th factor equal to 1 if the sink bits agree and equal to 0 if they differ. A nonzero shift is not killed by unitarity alone. Shifted distinct-row overlaps need not vanish. The process function is what forces the dangerous shift to vanish.

If $s = s'$, every δ_k is zero, because a bit XOR itself is zero. Every factor is a row norm, hence 1. The product of ones is 1. If $s \neq s'$, the lemma supplies a party k_* , not necessarily the head, with $\delta_{k_*} = 0$ and $s_{k_*} \neq s'_{k_*}$. That factor is the inner product of two distinct rows, hence 0. A product with a zero factor is zero, wherever the factor sits. If the zero is at the head, the product is 0 times the tail. If the zero is in the tail, the tail product is 0, and the head times 0 is 0. The head-only lemma is the first case. It is not the theorem.

Therefore

$$(MM^\dagger)_{s,s'} = \delta_{s,s'}. \quad (13)$$

Equal finite dimension gives $M^\dagger M = I$ as well, so M is unitary. Fixing the source at the identity recovers w . The compiled statement takes the disagreement witness as a hypothesis and concludes that the gram entry is 1 on the diagonal and 0 when a witness exists. The file that produces the witness is the disagreement lemma. The two halves meet at the index k_* .

Unitarity of the local operators does not kill the shift by itself. The process-function clause does. That is why the swap and the identity loop fail the same contraction. Both satisfy the identification. Neither has a witness party of the required kind for every unequal pair, and the gram of either one against a unitary local family is not the identity.

13. A walk through one unequal pair

The identification is easier to see on two parties than in the induction. Let $s = 00$, $s' = 10$, and suppose $w(00) = 01$ and $w(10) = 01$. The two strings differ in the first bit and agree in the value of w . Party 0 is a witness. The shift is $w(10) \oplus w(00) = 00$. The factored gram therefore has, at party 0, the inner product of row 0 with row 1, unshifted. If the rows are orthonormal, that factor is 0, and the product is 0, whatever the second party is doing. The same pair with the witness moved to party 1 puts the zero in the second factor. The head-only argument misses it. The list induction does not.

Now change the map so that $w(00) = 00$ and $w(10) = 11$. The strings still differ, but they differ in w wherever they differ as strings. There is no witness. The shifts are both 1. Each factor is a shifted overlap, and orthonormality does not say that a shifted overlap vanishes. This map is not a process function. The gram is not obliged to be the identity.

14. The copy rule and the swap, as controls

Take $w(o) = o$. The extension $U(o, e) = (o \oplus e, o)$ is still a permutation. The matrix element is still a product. The factoring lemma still applies, and the reindexing still produces $\delta_k = s'_k \oplus s_k$. On the diagonal every shift vanishes and the gram entry is 1, if the rows are orthonormal. Off the diagonal, a party with $s_k \neq s'_k$ has shift 1, not 0. The lemma's conclusion is false, because the hypothesis is false. For the one-bit matrix

$$H = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

the overlap of row 0 with row 1 shifted by 1 is $1 \cdot 1 + 1 \cdot 1 = 2$, not 0. The gram is not the identity. The identification held. Unitarity failed.

The swap is the other control. On two bits, $w(o_0, o_1) = (o_1, o_0)$. The strings 01 and 10 are distinct, and w agrees on them at both positions, so the shift is 00. Both factors are unshifted overlaps of distinct rows. Either one kills the product, if the rows are orthonormal. But the swap is not a process function, and other unequal pairs do not have a witness. The pair 00 and 11 has shift 11, and the gram entry need not vanish. A single vanishing pair is not unitarity. The definition fails globally, and so does the gram.

15. What was checked

Statement	Where it stands
If $s \neq s'$, some party disagrees in the string and agrees in w .	Checked for every finite number of parties on bits.
Both compositions of the open shuffle.	Checked on three bits. XOR cancels coordinatewise in general.
Sum of a product over $\{0, 1\}^n$ equals the product of the sums.	Checked, integer entries, arbitrary party list.
XOR with a fixed bit preserves the two-point sum.	Checked.
The source sum equals the product of shifted row overlaps.	Checked.
Orthonormal rows and $s = s'$ give gram entry 1.	Checked.
A witness at any index gives gram entry 0.	Checked. The head-only case is not the theorem.
Closed contraction, copy rule, three bits.	Gram diagonal 8, not 1.

Complex conjugation.	An involution on Gaussian integers. Reindexing does not move it.
A named Einstein metric for Lugano.	Not constructed.

The integer file is the real case of the same algebra. The ring steps actually used are commutativity and associativity of multiplication, distributivity, and commutativity of addition for the swap in the reindexing. Those hold in $\mathbb{C}$. Checking them on $\mathbb{Z}$ does not construct a complex unitary. It checks the cancellation that the complex argument needs.

16. Instruments, at the width that is actually a theorem

A local instrument is a completely positive trace-preserving map on the party's system. Finite-dimensional Stinespring says that every such map is a unitary on the system together with an ancilla, the ancilla prepared in a fixed state, and a partial trace. If operators E_i satisfy $\sum_i E_i^\dagger E_i = I$, the map

$$V|\psi\rangle = \sum_i |i\rangle \otimes E_i|\psi\rangle \quad (14)$$

is an isometry, and the channel is the partial trace of the conjugation by V . Extending V to a unitary on a larger ancilla is a choice of the unused columns. That choice is not unique, and it is not needed for the partial trace.

Prepare the ancilla in $|0\rangle$ and let the process function act on the system bits only. The columns indexed by ancilla input 0 are a subset of the columns of a unitary, so they are orthonormal. That inheritance is local. It does not cancel the closed contraction. Section 7 is the counterexample: against local identities the copy rule has gram diagonal 2^n . The open step, the bijection followed by the local isometry, has not fed the output back. Closing it is a different map, and it is not an isometry for every rule.

The partial trace of an isometry is completely positive and trace-preserving. That sentence is the instrument step for the open dilation. The process-function clause is not what makes that open step trace-preserving. It is what makes the no-ancilla contraction unitary, and what makes source 0 recover a uniquely completing w .

17. Conjugation, as a second file

The integer file treats the bar as the identity. The second file does not. A Gaussian integer is a pair of integers, and

$$\overline{a + bi} = a - bi \quad (15)$$

is the compiled map. It is an involution, a ring homomorphism, and it sits on the second factor of the overlap. XOR reindexing does not move it: conjugating after the swap of summands equals swapping after conjugating. A zero factor still kills the product. The file does not claim the closed isometry for every rule, and it does not define a metric.

18. What this does not decide

A process function does not need a metric, because the definition never mentions a distance, a time, or a line element. Each party sends a bit and receives a bit. The function w takes the tuple of sent bits to the tuple of received bits, and it is admitted only when every local rule finishes exactly once. The parties are labels. They are not points of a manifold. The source-and-sink map $U(o, e) = (w(o) \oplus e, o)$ is a permutation of bit strings. XOR is addition modulo 2, not a displacement along a curve. The cancellation that makes the closed map unitary is an inner product of rows of a matrix on a bit, not the inner product of a spacetime. A metric would be needed if the claim were that these parties are events on a Lorentzian manifold and that w is the scattering of a field along a closed timelike curve. That claim is not made. A metric would be a realization, not a missing premise.

The theorem does not embed the process in a spacetime. A process function is a constraint on maps between parties. A closed timelike curve is a constraint on the causal structure of a Lorentzian manifold. The essay has not shown that every process function arises as the scattering map of a field on a spacetime with a closed curve, and it has not shown the converse. Gödel's solution and the wormhole construction remain existence proofs in a different theory [10], [11]. The billiard remains an ill-posed classical initial-value problem [12]. Nothing here assigns a metric to Lugano.

The classical literature already has a picture, and it should not be inflated into a metric. Implement the function by collisions: a bit is the absence or presence of a ball, and walls or further balls compute a function of the incoming balls. Route each party's output, through a wormhole or another closed curve, to the input of the collision region. That picture is what the 2019 paper records as a realisation of the reversible extension [3]. It is not a solution of Einstein's equation written down here. The wormhole needs stress-energy that violates the averaged null energy condition [11]. The process function is the selection rule that admits only the uniquely completing collisions. A named Einstein metric for Lugano was not added.

The theorem does not decide whether the agent had a choice. Lewis's two cans are still two cans [1]. The process function says which global maps complete uniquely. It does not say, from inside one party's laboratory, whether the intervention that was not the fixed point was an action the party could have taken. Black's demand is met only in the thin sense that the failed bilking is not a fixed

point of the admitted map [8]. That is a dynamics, of a discrete and deterministic kind. It is not a story about freedom.

The theorem does not say that nature uses process functions. It says that if a deterministic multi-party map completes every local intervention uniquely, then its source-and-sink extension, contracted against local unitaries with orthonormal rows, is a unitary from source to sink. Unique completion was already the condition that excluded the grandfather and the bootstrap. The contraction shows that the same condition is the vanishing of the off-diagonal. The forbidden double fixed point is the forbidden gram entry. Lugano is an instance, not the argument.

19. Why the fixed-point clause is the whole condition

It is tempting to read the theorem as a quantization recipe: start with a classical consistent process, add a source and a sink, contract, and obtain a quantum process. The recipe is accurate, and it is easy to overread. The consistency that went in was already a uniqueness statement about interventions. The unitarity that came out is a uniqueness statement about amplitudes: the source-to-sink map sends an orthonormal basis to an orthonormal basis. The proof says these are the same uniqueness, transported across the extension. A second consistency principle, imposed on the quantum side as a measure-zero axiom or as a post-selection, would be doing the job a second time.

Deutsch's fixed point is a different uniqueness, and a weaker one [15]. It asks for a state on the curve that reproduces itself after the partial trace. Many interactions have such a state, including interactions that a process function would reject, and including, as Tolktsdorf and Verch showed, evolutions with no curve at all [18]. Post-selection is a different uniqueness again [19]. It asks that the retained sample be the one in which the loop closed. A process function asks that the loop close, once, for every local choice, before any sample is drawn. The contraction is the reason that last demand is already a quantum demand.

None of this makes the classical billiard well posed. The ball in the wormhole spacetime has, for some data, two consistent continuations, and a process function would refuse the analogous multi-party map. The refusal is not a solution of the differential equation. It is a reason to think that the map induced by that data is not the sort of map the present theorem quantizes. Earman's warning still applies to any theory that selects after solving [14]. It applies less directly to a theory whose admitted maps were defined to be the uniquely completing ones. Whether general relativity has a formulation of that kind is not settled here.

20. Bilking, written as an empty preimage

Black's argument can be written without the word "can" [8]. Let the earlier event be a record r , and the later event an action a . A bilking intervention is a local function that reads r and returns an action incompatible with the action the process would require in order to have produced r . In the notation of the fixed-point equation, the intervention f asks for an output o with $o = f(w(o))$, and the bilking choice of f is one for which no such o exists. Novikov's principle says that this f is not among the interventions that occur, or that the corresponding history has measure zero [2]. A process function says something prior to occurrence: a map that admits this f is not a process function. The empty preimage is the inconsistency. There is nothing further to set to measure zero.

The bootstrap is the dual emptiness. An intervention with two solutions has a preimage that is too large. Information "from nowhere" is a misleading description of that fact. The note in the coat does not appear from nowhere in the fixed-point equation. It appears twice, which is already too often. Lewis's contextual can fits here without being the criterion [1]. Relative to the local form of f , both writings are possible. Relative to the admitted map, an f with two fixed points is not an intervention the map supports. The context shift is real. The dynamics is the restriction on w .

The finite-abelian version is the same essay with a larger alphabet. Replace the bit by a finite abelian group, XOR by the group law, and the two-point sum by the sum over the group. Translation by a fixed element remains a bijection, so the reindexing step is unchanged, and the shift is a difference rather than an XOR. The disagreement lemma needs one extra check: the local function that sends $w_k(s)$ to s_k and $w_k(s')$ to s'_k is well defined when those inputs coincide only if the outputs coincide, which is the same supposition as on bits. The compiled file is the bit case. The group case is the same induction with a larger finite sum.

21. On the stories

The clerk and the note are not models. They are there so that the two failures of the fixed-point equation have a face before they have an index. Mara's closed window is an intervention with no solution. The note in the coat is an intervention with two. The third telling, in the witness section, is the lemma itself: two visits, one reply, a line that differs and a line that does not. A reader who finds the stories distracting can skip them. Nothing in the proof cites them. A reader who finds the indices distracting can keep the clerk, and then come back to party k_* , which is the line on the form.

One temptation is worth refusing. It is possible to tell the stories so that the loop "wants" to be consistent, or so that the office "knows" the future. The definition does not say that. A process function is a set of completions. It has no attitude toward the completions it lacks. The unitary lift has no attitude either. It is a matrix that happens to be unitary precisely when the set of completions was already unique.

What remains open is not a missing line of this argument. It is the distance between a process function and a spacetime. A reader who wants the clerk to be a curve in a Lorentzian manifold wants a theorem this essay does not contain. A reader who wants the note to have been written freely wants a theorem about agency, which the fixed-point equation is not. The proved statement is smaller, and it is the one the contraction supports. Every process function lifts, by the source-and-sink permutation, to a source-to-sink unitary, once the local matrices have orthonormal rows. The lift uses nothing but the reason the process function was consistent.

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Draft. Not a preprint. The stories are the two failures and the witness. They are not evidence. The open shuffle is not the closed contraction.